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S. No. of Question Paper : 2254

Unique Paper Code : 32351602

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Name of the Paper : Ring Theory and Linear Algebra- II

Name of the Course : B.Sc. (Hons.) Mathematics

Semester : VI

Duration : 3 Hours

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

Attempt any two parts from each of the questions.

1. (a) Let $f(x) = x^3 + 2x + 4$ and $g(x) = 3x + 2$ in $\mathbb{Z}_5[x]$. Determine the quotient and remainder upon dividing $f(x)$ by $g(x)$.
(b) State and prove Einstein's criterion of irreducibility.
(c) If D is a principal ideal domain, prove that every strictly increasing chain of ideals $I_1 \subset I_2 \subset I_3 \dots$ must be finite in length. Hence, prove that every non-zero and a non-unit element of D has an irreducible factor. 6.5, 6.5, 6.5
2. (a) Prove that the ideal $\langle x \rangle$ in $\mathbb{Q}[x]$ is maximal.
(b) (i) Prove that $8x^3 - 6x + 1$ is irreducible over $\mathbb{Q}$.
(ii) Prove that the product of two primitive polynomials is primitive.
(c) Prove that in a principal ideal domain, an element is irreducible if and only if it is prime. 6,6,6

P.T.O.

3. (a) Suppose that V is a finite dimensional vector space with the ordered basis $\beta = \{x_1, x_2, \dots, x_n\}$. Let $f_i (1 \leq i \leq n)$ be the i th coordinate function with respect to β be defined such that $f_i(x_j) = \delta_{ij}$ where δ_{ij} is the Kronecker delta. Let $\beta^* = \{f_1, f_2, \dots, f_n\}$. Then prove that β^* is an ordered basis for V^* , and, for any $f \in V^*$, we have
- $$f = \sum_{i=1}^n f(x_i) f_i.$$

- (b) For $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \in M_{2 \times 2}(\mathbf{R})$, determine the eigen values

of A and eigen space corresponding to each eigen value of A . Also, if possible, find a basis for $\mathbf{R}^2$ consisting of eigen vectors of A .

- (c) Prove that the characteristic polynomial of any diagonalizable linear operator splits. Is the converse true ? Justify. 6.5, 6.5, 6.5

4. (a) Let T be a linear operator on $\mathbf{R}^3$ such that :

$$T(a, b, c) = (a + b + c, a + b + c, a + b + c).$$

Let $W = \{(t, t, t) \mid t \in \mathbf{R}\}$ be a subspace of $\mathbf{R}^3$. Show that :

- (i) W is a T -invariant subspace of $\mathbf{R}^3$.
 - (ii) The characteristic polynomial of T_W divides the characteristic polynomial of T .
- (b) Let D be the differentiation operator on $P(\mathbf{R})$, the space of polynomials over $\mathbf{R}$. Prove that there exists no polynomial $g(t)$ for which $g(D) = T_0$. Hence, show that D has no minimal polynomial.

- (c) Let T be a linear operator on a finite dimensional vector space V and let $p(t)$ be the minimal polynomial of T . Prove that a scalar λ is an eigen value of T if and only if $p(\lambda) = 0$. 6, 6, 6

5. (a) Let V be an inner product space. Prove that :

(i) $\|x \pm y\|^2 = \|x\|^2 \pm 2\operatorname{Re}\langle x, y \rangle + \|y\|^2$ for all $x, y \in V$
 where $\operatorname{Re}\langle x, y \rangle$ denotes the real part of the complex number $\langle x, y \rangle$.

(ii) $|\|x\| - \|y\|| \leq \|x - y\|$ for all $x, y \in V$.

(b) Suppose that $S = \{v_1, v_2, \dots, v_k\}$ is an orthonormal set in an n -dimensional inner product space V . Show that :

(i) S can be extended to an orthonormal basis $\{v_1, v_2, \dots, v_k, v_{k+1}, \dots, v_n\}$ for V .

(ii) If $W = \operatorname{span}(S)$ then $S_1 = \{v_{k+1}, \dots, v_n\}$ is an orthonormal basis for $W^\perp$.

(iii) If W is any subspace of V , then $\dim V = \dim W + \dim W^\perp$.

- (c) Find the orthogonal projection of the given vector on the given subspace W of the inner product space :

$$V = \mathbb{R}^3, u = (2, 1, 3) \text{ and}$$

$$W = \{(x, y, z) : x + 3y - 2z = 0\}. \quad 6.5, 6.5, 6.5$$

6. (a) State and prove Bessel's inequality.
- (b) For the inner product space V over F and linear transformation $g : V \rightarrow F$, find a vector y such that $g(x) = \langle x, y \rangle$ for all $x \in V$ where :

$$V = P_2(\mathbb{R}) \text{ with } \langle f, h \rangle = \int_0^1 f(t)h(t)dt, \text{ and}$$

$$g(f) = f(0) + f'(1).$$

- (c) Let V be a complex inner product space and let T be a linear operator on V .

$$\text{Define } T_1 = \frac{1}{2}(T + T^*) \text{ and } T_2 = \frac{1}{2i}(T - T^*)$$

- (i) Prove that T_1 and T_2 are self adjoint and that $T = T_1 + iT_2$.
- (ii) Suppose also that $T = U_1 + iU_2$ where U_1 and U_2 are self-adjoint. Prove that $U_1 = T_1$ and $U_2 = T_2$.
- (iii) Prove that T is normal if and only if $T_1T_2 = T_2T_1$.

6,6,6